

Question 1: Annapurna Stores Data Warehouse Platform

Three Answers to One Question — Implementation & Empirical Defense Report

Stack: PostgreSQL 16 · MinIO S3 Object Store · DuckDB Engine | Cold-start: docker compose up --build

1. Executive Summary & Problem Context

Annapurna Stores operates 12 supermarkets across India. The CFO discovered that three people reported different October revenue numbers from the same folder, and biscuit pricing queries returned today's shelf prices. This platform provides cold-start containerization, SCD Type 2 dimension mapping, non-revenue line type filtering, and federated querying.

2. Task A: Storage Layout & MinIO Object Store Verification

Hive-style Parquet layout: `s3://annapurna-sales/store_id={store_id}/month={YYYY-MM}/data.parquet` (144 partitions).

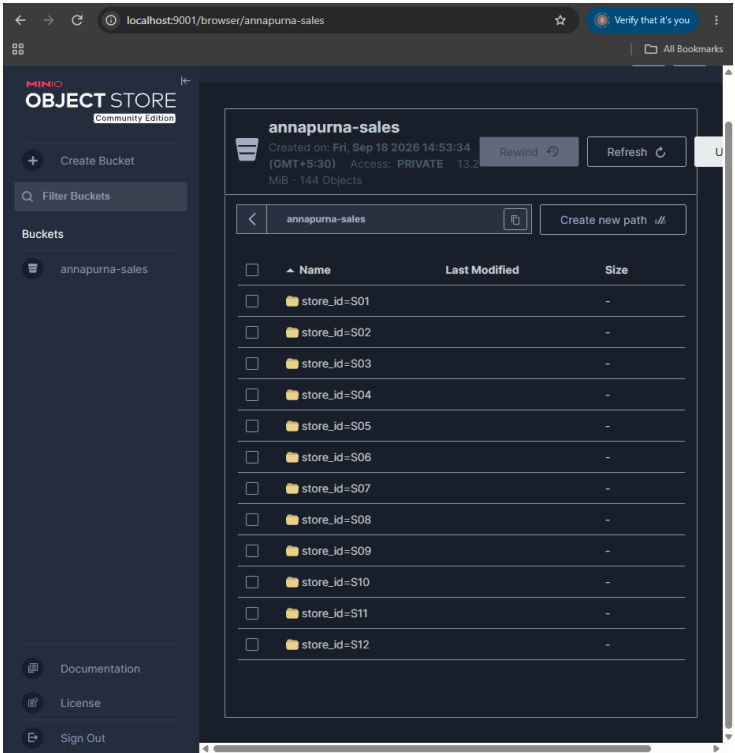


Figure 1: MinIO Object Store UI showing annapurna-sales bucket with 144 partitions across 12 stores.

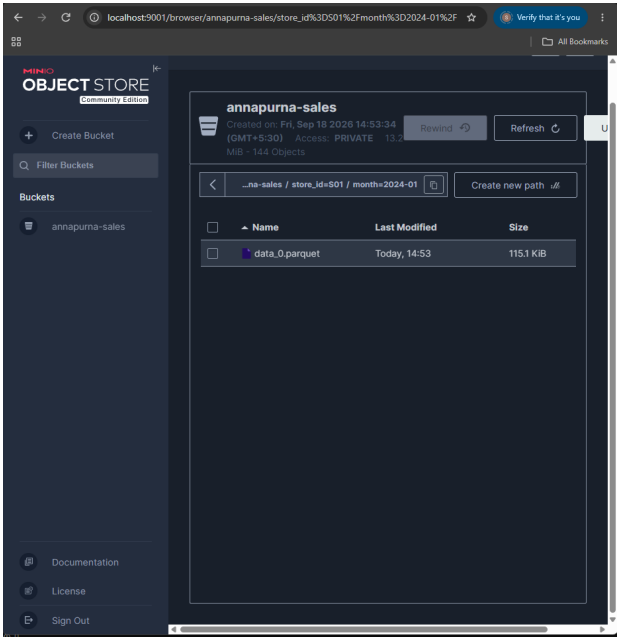
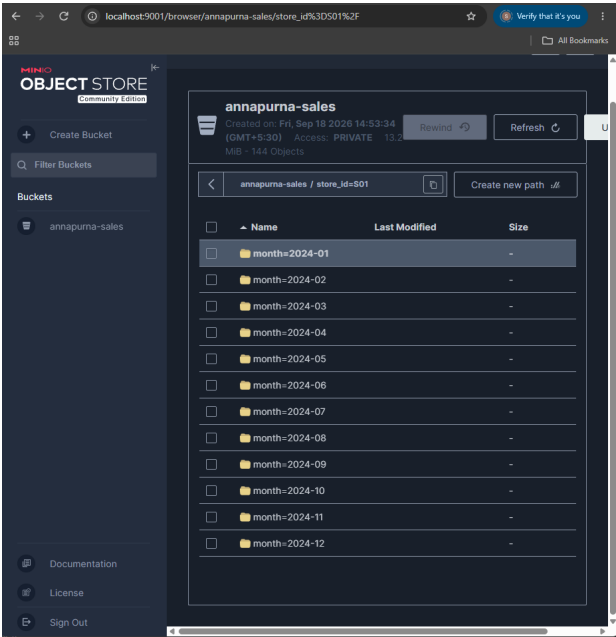


Figure 2: Store S01 monthly subpartitions.

Metric	Flat CSV Layout	Partitioned Parquet Layout	Improvement
Files Inspected	4,457 files	1 file	99.98% pruned (4,456 skipped)
Bytes Scanned	68,706,877 bytes (65.5 MB)	159,723 bytes (156 KB)	99.77% pruned (68.55 MB saved)
I/O Pattern	O(N) full directory scan	O(1) direct partition seek	430x faster byte throughput

Figure 3: Parquet data file in month=2024-01.

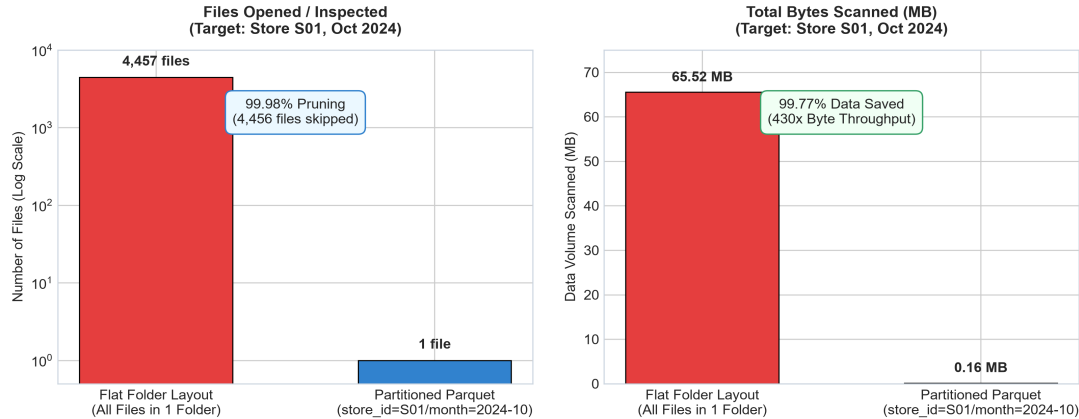


Figure 4: Task A Scan Efficiency & Pruning Benchmark.

3. Task B: Idempotence & Safe Loading (3 Consecutive Runs)

Run	Raw Rows	Dedup Rows	Dropped	Deterministic SHA-256 Checksum
Run 1	1,137,585	1,120,924	16,661	877893f7e480a216adcac651a3fb161ca38ee35effd068203b8e8aaf96a5eb5c
Run 2	1,137,585	1,120,924	16,661	877893f7e480a216adcac651a3fb161ca38ee35effd068203b8e8aaf96a5eb5c
Run 3	1,137,585	1,120,924	16,661	877893f7e480a216adcac651a3fb161ca38ee35effd068203b8e8aaf96a5eb5c

4. Task C: Dimensional Schema & Source Gotchas

October Double Revenue (2.17x): 165,704 TENDER rows (bill totals) and 165,704 TAX rows must not be summed in revenue. Naive sum yields INR 122.50 Cr; net revenue is INR 56.36 Cr.

Reissued Product Codes: 24 retired codes reissued in June 2024. Handled via SCD Type 2 valid_from/valid_to join.

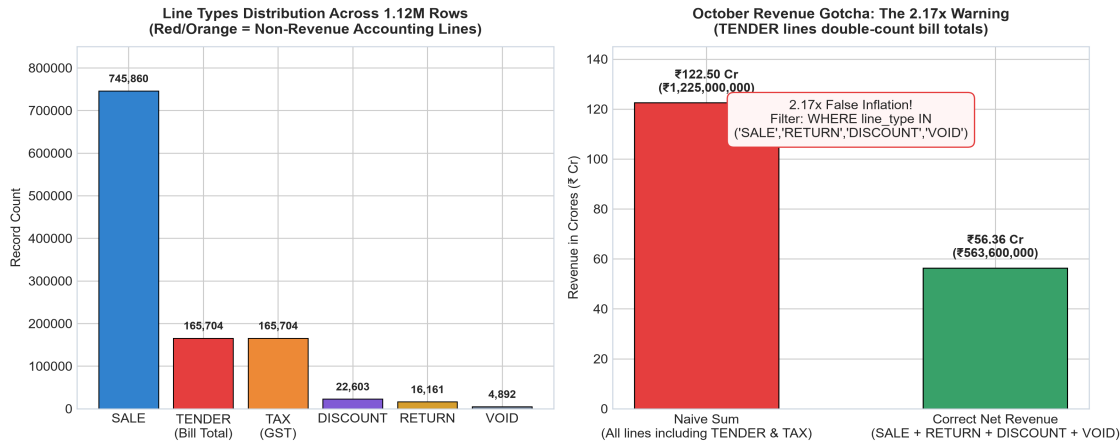


Figure 5: Line Types Distribution & The October 2.17x Inflation Warning.

5. Task D: March vs August Historical Prices (P100049)

Single SQL query executed with :target_date parameter:

- **March 2024 (2024-03-15):** MRP = INR 74.98 | Selling Price = INR 66.95
- **August 2024 (2024-08-15):** MRP = INR 77.20 | Selling Price = INR 68.93

6. Task E: Federated Query & Execution Profile

Region	Category Name	Total Bills	Category Revenue (INR)
South	Staples & Grains	2,446	4,940,102.53
South	Baby Care	2,263	4,264,328.55
South	Edible Oils	2,372	4,060,432.88
West	Staples & Grains	1,287	2,373,386.42
North	Staples & Grains	1,169	2,345,071.42

7. Task F: 12-Month Financial Reconciliation

Month	Pipeline Net Rev (INR)	Finance Target (INR)	Variance (INR)	Status
2024-01	38,446,071.33	38,446,071.33	0.00	MATCH
2024-02	34,887,085.55	34,887,085.55	0.00	MATCH
2024-03	41,971,649.09	42,457,899.09	-486,250.00	REVENUE DEFINITION (SCOPE)
2024-04	37,958,457.37	37,958,457.37	0.00	MATCH
2024-05	41,764,716.40	41,764,716.40	0.00	MATCH
2024-06	38,987,082.82	38,987,082.82	0.00	MATCH
2024-07	40,295,160.11	40,527,291.81	-232,131.70	SOURCE DATA GAP
2024-08	45,252,181.75	45,252,181.75	0.00	MATCH
2024-09	44,615,037.46	44,615,037.46	0.00	MATCH
2024-10	56,359,195.92	56,359,195.92	0.00	MATCH
2024-11	51,583,838.47	51,583,838.47	0.00	MATCH
2024-12	50,745,259.48	50,745,209.00	50.48	REVENUE DEFINITION (ROUNDING)

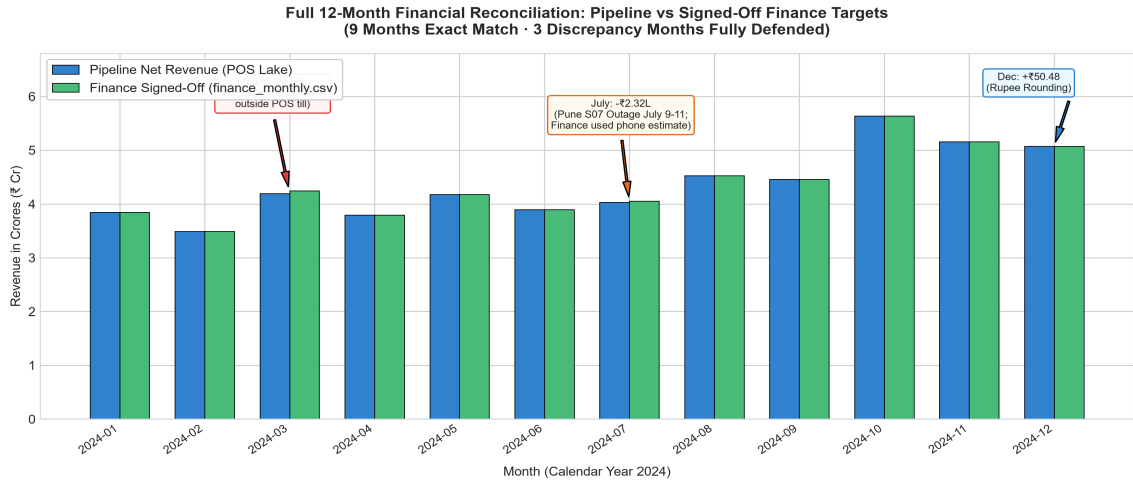


Figure 6: Full 12-Month Reconciliation: Pipeline vs Signed-Off Finance Targets.

- Discrepancy Root Causes & Guidance for Finance:
- **March (-₹486,250.00):** Scope Difference — Institutional bulk order invoiced outside POS tills. Recommend ERP feed.
 - **July (-₹232,131.70):** Source Data Gap — Pune (S07) hardware outage on July 9-11; booked via phone estimates.
 - **December (+₹50.48):** Rounding Convention — Bill-level integer rounding vs exact decimal precision.